

PENRHYN, Cook Islands

WMO: 918020

Lat: 9.017S

Lon: 158.031W

Elev: 3

StdP: 14.69

Time Zone: -10.00 (W10)

Period: 86-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	76.2	77.0	68.4	104.4	81.7	69.7	109.2	82.0	29.0	79.5	25.3	80.4	8.5	80	0.533

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	5.9	91.0	80.4	90.1	80.1	89.4	79.8	82.1	88.1	81.6	87.5	81.1	86.8	6.4	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.5	158.6	86.3	79.9	155.2	85.8	79.3	151.8	85.2	46.0	88.7	45.4	87.8	44.8	87.5	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	22.5	20.7		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	83.2	83.2	83.0	83.5	83.8	83.8	83.4	83.1	82.6	82.8	82.7	83.1	83.2
	DBStd	1.77	1.50	1.90	2.04	1.66	1.65	1.49	1.61	1.50	1.64	1.65	1.76	2.19
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12110	1028	924	1038	1013	1047	1002	1026	1010	985	1014	994	1029
	CDD65	6635	563	504	573	563	582	552	561	545	535	549	544	564
	CDH74	78125	6666	5905	6874	6870	7207	6545	6442	6111	6127	6361	6426	6591
	CDH80	26788	2287	1990	2539	2607	2782	2301	2075	1779	1922	2023	2189	2294
(14) Wind	WSAvg	11.1	12.1	11.8	9.9	10.2	12.5	12.3	10.5	11.5	12.5	12.0	9.4	9.1
(15) Precipitation	PrecAvg	56.3	7.8	7.2	7.1	4.1	3.5	3.3	3.7	3.2	2.5	3.9	3.8	6.1
	PrecMax	98.9	19.4	27.2	15.4	14.4	9.3	9.8	10.5	10.5	5.2	7.6	11.3	17.8
	PrecMin	20.2	1.7	1.1	1.5	1.3	0.6	0.8	0.6	0.6	0.4	0.5	0.8	0.5
	PrecStd	18.5	5.0	6.2	4.0	2.8	2.2	2.0	2.8	2.6	1.3	1.8	2.4	4.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.2	90.8	90.8	90.9	91.7	90.1	90.6	90.0	90.7	91.3	91.2	91.7
		MCWB	79.9	81.2	81.7	81.3	80.5	80.0	79.4	79.2	79.2	80.9	79.5	80.3
	2%	DB	89.3	89.3	89.5	89.6	89.8	88.9	89.1	88.6	88.9	89.4	89.6	90.2
		MCWB	79.1	80.5	81.1	80.8	80.1	79.5	79.1	78.6	78.5	79.2	79.3	80.1
	5%	DB	87.8	87.8	88.2	88.2	88.2	87.5	87.7	87.2	87.7	87.7	88.2	89.0
		MCWB	78.9	80.0	80.5	80.4	79.7	79.3	78.5	78.0	78.1	78.6	78.9	79.8
	10%	DB	86.3	86.3	86.9	87.0	86.7	86.3	86.3	85.9	86.1	86.2	86.9	87.6
		MCWB	78.8	79.3	79.9	80.1	79.4	78.9	78.1	77.6	77.9	78.3	78.6	79.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.5	82.4	82.8	82.8	82.0	81.7	81.1	80.9	81.0	81.7	81.8	82.1
		MCDB	87.8	89.3	89.0	88.2	87.6	87.5	88.0	86.1	86.5	89.1	87.6	89.0
	2%	WB	80.7	81.3	81.9	81.9	81.2	80.9	80.1	80.0	80.1	80.6	80.9	81.0
		MCDB	86.4	87.6	87.8	87.2	86.7	86.4	86.5	85.4	85.8	87.2	86.7	87.9
	5%	WB	80.0	80.6	81.1	81.4	80.8	80.2	79.5	79.4	79.5	79.9	80.1	80.4
		MCDB	85.5	86.5	86.9	86.7	86.2	85.6	85.5	84.7	85.0	85.8	85.9	87.1
	10%	WB	79.6	79.9	80.6	80.8	80.3	79.7	79.0	78.9	79.0	79.2	79.5	79.8
		MCDB	84.9	85.3	86.1	86.0	85.6	85.0	84.6	84.1	84.3	84.6	85.1	85.9
(35) Mean Daily Temperature Range	5% DB	MDBR	6.1	6.2	6.6	6.1	5.9	6.3	6.3	6.3	6.6	6.1	6.5	7.1
		MCDBR	9.4	8.6	8.6	8.2	8.3	8.1	8.9	8.9	9.4	9.3	9.2	9.7
		MCWBR	4.2	4.0	4.2	3.9	3.8	3.8	4.2	4.1	3.9	4.0	3.9	4.2
	5% WB	MCDBR	6.7	7.3	7.4	6.7	6.2	6.7	7.0	6.6	6.8	7.3	6.9	8.3
		MCWBR	3.7	3.9	4.2	4.0	3.8	3.7	3.6	3.9	3.9	4.0	3.7	4.3
(40) Clear-Sky Solar Irradiance	taub		0.383	0.386	0.381	0.384	0.384	0.386	0.384	0.381	0.390	0.379	0.378	0.382
	taud		2.625	2.615	2.633	2.635	2.634	2.604	2.597	2.596	2.568	2.606	2.621	2.615
	Ebn at Noon		304	301	298	288	279	273	276	285	291	300	303	303
	Edh at Noon		32	32	31	30	29	29	29	30	33	32	32	32
(44) All-Sky Solar Radiation	RadAvg		1917	1934	1893	1848	1685	1605	1680	1879	2055	2092	2091	2007
	RadStd		192	246	166	128	96	95	120	93	97	121	127	211

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page